

Frontend Architecture

React, routing, auth, components, and API integration guide

1. Project Structure

```
FrontEnd/  
  ■■■ public/  
    ■ ■■■ index.html # Root HTML file  
  ■■■ src/  
    ■ ■■■ api/  
      ■ ■ ■■■ axios.js # Axios instance + JWT interceptor + 401 redirect  
    ■ ■■■ context/  
      ■ ■ ■■■ AuthContext.js # Global auth state (login, register, logout, user)  
    ■ ■■■ components/  
      ■ ■ ■■■ Layout.js # Sidebar + AppBar + for nested routes  
    ■ ■■■ pages/  
      ■ ■ ■■■ Login.js # Login form with JWT flow  
      ■ ■ ■■■ Register.js # Registration with password strength meter  
      ■ ■ ■■■ Dashboard.js # Overview: wallets, charts (Area, Pie, Bar)  
      ■ ■ ■■■ Exchange.js # Currency converter + rate chart + history  
      ■ ■ ■■■ Transfer.js # UPI P2P transfer form + stepper + history  
      ■ ■ ■■■ NFTMarketplace.js # Mint, browse, buy NFTs (card grid + dialogs)  
      ■ ■ ■■■ NachaPayment.js # ACH payment form + flow visualization + history  
      ■ ■ ■■■ Account.js # Profile + wallets + add currency wallet  
    ■ ■■■ App.js # BrowserRouter + Routes + ProtectedRoute  
    ■ ■■■ index.js # ReactDOM.render entry point  
    ■ ■■■ index.css # Global styles (dark background)  
  ■■■ .env # REACT_APP_API_URL=http://localhost:8080/api  
  ■■■ package.json # Dependencies
```

2. Authentication Flow

Authentication uses React Context API for global state and localStorage for token persistence. The JWT is stored in localStorage and automatically attached to every API request via an Axios request interceptor. A 401 response from the server clears the token and redirects to /login.

■ Login Flow

- User enters username + password on Login.js
- AuthContext.login() calls POST /api/auth/login
- Server validates credentials, returns { token, username, email, role }
- Token saved to localStorage.token, user object to localStorage.user
- user state updated in AuthContext → React re-renders
- App.js ProtectedRoute sees user → redirects to /dashboard

■ Register Flow

- User fills Register.js form (username, email, phone, password)
- Password strength meter validates: 8+ chars, contains letter, contains number
- AuthContext.register() calls POST /api/auth/register
- Server creates user + 3 wallets (INR ₹10,000, USD \$0, EUR €0)
- Returns JWT immediately — user is logged in after registration
- Redirected to /dashboard showing new wallets

■ Axios Interceptors

```
// Request interceptor – attach JWT
api.interceptors.request.use(config => {
  const token = localStorage.getItem('token');
  if (token) config.headers.Authorization = `Bearer ${token}`;
  return config;
});

// Response interceptor – handle 401
api.interceptors.response.use(
  res => res,
  err => {
    if (err.response?.status === 401) {
      localStorage.removeItem('token');
      localStorage.removeItem('user');
      window.location.href = '/login'; // Hard redirect
    }
    return Promise.reject(err);
  }
);
```

3. Routing Structure

```
} /> { /* Public */}
} /> { /* Public */}

}>
```

```
    } />
  } />
} />
} />
} />
} />
} />
} />
} />
```

ProtectedRoute checks AuthContext.user — if null (not logged in), it redirects to /login. If user exists, it renders children (Layout, which renders — the matched child route page).

4. Charts & Visualizations

Page	Chart Type	Library	Data Source
Dashboard	AreaChart — Balance History	Recharts	Simulated 6-month data (replace with API)
Dashboard	PieChart — Portfolio Split	Recharts	Wallet balances from /users/{id}/wallets
Dashboard	BarChart — Monthly Activity	Recharts	Simulated (replace with activity API)
Exchange	LineChart — Rate Today	Recharts	Simulated intraday rate with sin wave

5. Material UI Dark Theme

The entire application uses a custom dark theme with gold (#FFD700) as the primary accent color. The background is a deep dark blue-black gradient (#07071a to #0d0d22). Cards use a glassmorphism effect (semi-transparent background with backdrop blur).

■ Design System Colors

Variable	Hex	Usage
Gold	#FFD700	Primary actions, active nav, wallet INR, stat values
Dark BG	#07071a → #0d0d22	Main page background gradient
Card BG	rgba(255,255,255,0.04)	Card backgrounds (glassmorphism)
USD Green	#00C49F	USD wallet, success states
EUR Blue	#0088FE	EUR wallet, info states
GBP Orange	#FF8042	GBP wallet, warning states
JPY Purple	#a855f7	JPY wallet
Border	rgba(255,255,255,0.08)	Card borders, dividers

6. Key npm Dependencies

Package	Version	Purpose
react	18.x	Core UI library
react-router-dom	6.x	Client-side routing
@mui/material	5.x	Component library
@mui/icons-material	5.x	Material Design icons
recharts	2.x	Chart visualizations
axios	1.x	HTTP client
@emotion/react	11.x	MUI styling engine
@emotion/styled	11.x	MUI styled components